

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

As a data scientist, you must develop a model which will predict the insurance charges.

1.) Identify your problem statement :

- **Stage 1-** Machine Learning because dataset in numbers.
- **Stage 2** – Supervised Learning because Requirement is clear in the dataset and output in number.
- **Stage3** – Regression because we going to predict the insurance charges column this output are in numbers.

2.) Tell basic info about the dataset (Total number of rows, columns) :

- This Dataset contains **1338 rows × 6 columns** and name of column are ['age', 'sex', 'bmi', 'children', 'smoker', 'charges']

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

- Column **Sex[Male, Female]** and **Smoker[Yes,No]** convert this categorical column into numerical by **One-Hot encoding technique** into separate column **male** and **female** **smoker yes** and **smoker no** with value **0** and **1**.

4.) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Below models I have used in this Problem statement

- Multiple LinearRegression
- Support Vector Machine Regression
- Decision Tree Regression
- Random Forest Regression

5.) All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

- 1) **Multiple LINEAR REGRESSION (R2 Value) = 0.7681**
- 2) **SUPPORT VECTOR MACHINE:**

S.no	Hyper Parameter	Linear (r2_score)	RBF (Non Linear r2_score)	Poly (r2_score)	Sigmoid(r2 Score)
1	C0.01	-0.0570	-0.083	-0.083	-0.083
2	C10	0.1265	-0.076	-0.0885	-0.082
3	C100	0.5965	-0.114	-0.108	-0.1079
4	C1000	0.7432	-0.091	-0.05377	-1.1951
5	C2000	0.7120	-0.091	0.01566	-4.2506
6	C5000	0.7004	-0.0303	0.2038	-23.586

- The **SVM Regression** use **R2 value**(Linear and hyper parameter (C1000) = **0.7432**)
- 3)DECISION TREE Regression

S.NO	criterion	MAX_FEATURE	SPLITTER	R VALUE
1	MSE	Auto	Best	0.74589
2	MSE	Auto	Random	0.6824
3	MSE	Sqrt	Random	0.71038
4	MSE	Sqrt	Best	0.69471
5	MSE	Log2	Random	0.71038
6	MSE	Log2	Best	0.69471
7	friedman_mse"	Auto	Best	0.74589
8	friedman_mse"	Auto	Random	0.682488
9	friedman_mse"	Sqrt	Random	0.71038
10	friedman_mse"	Sqrt	Best	0.69471
11	friedman_mse"	Log2	Random	0.71038
12	friedman_mse"	Log2	Best	0.69471
13	absolute_error	Auto	Best	0.77127
14	absolute_error	Auto	Random	0.7255
15	absolute_error	Sqrt	Random	0.63618
16	absolute_error	Sqrt	Best	0.757705
17	absolute_error	Log2	Random	0.63618
18	absolute_error	Log2	Best	0.757705
19	poisson	Auto	Best	0.76851
20	poisson	Auto	Random	0.688223
21	poisson	Sqrt	Random	0.71038
22	poisson	Sqrt	Best	0.66101
23	poisson	Log2	Random	0.71038
24	poisson	Log2	Best	0.66101

The **DECISION TREE Regression** use R2 value(Best hyper parameter is poisson,Auto,Best = **0.76851**)

- 4)Random Forest Regression

S.NO	criterion	MAX_FEATURE	n_estimators	R VALUE
1	MSE	Auto	100	0.8409
2	MSE	Auto	1000	0.83833
3	MSE	Auto	5000	0.84115
4	MSE	Sqrt	100	0.8466
5	MSE	Sqrt	1000	0.8477
6	MSE	Sqrt	5000	0.84822
7	MSE	Log2	100	0.8466
8	MSE	Log2	1000	0.8477
9	MSE	Log3	5000	0.84822
10	friedman_mse"	Auto	100	0.8383
11	friedman_mse"	Auto	1000	0.8409
12	friedman_mse"	Auto	5000	0.8411
13	friedman_mse"	Sqrt	100	0.8466
14	friedman_mse"	Sqrt	1000	0.84779
15	friedman_mse"	Sqrt	5000	0.8482
16	friedman_mse"	Log2	100	0.8466
17	friedman_mse"	Log2	1000	0.84779
18	friedman_mse"	Log3	5000	0.8482
19	absolute_error	Auto	100	0.83684
20	absolute_error	Auto	1000	0.83545
21	absolute_error	Auto	5000	0.8358
22	absolute_error	Sqrt	100	0.8495
23	absolute_error	Sqrt	1000	0.8494
24	absolute_error	Sqrt	5000	0.84927
25	absolute_error	Log2	100	0.8495
26	absolute_error	Log2	1000	0.8494
27	absolute_error	Log3	5000	0.84927
28	poisson	Auto	100	0.8412
29	poisson	Auto	1000	0.8421
30	poisson	Auto	5000	0.8429
31	poisson	Sqrt	100	0.8492
32	poisson	Sqrt	1000	0.84878
33	poisson	Sqrt	5000	0.8482
34	poisson	Log2	100	0.8492
35	poisson	Log2	1000	0.8487
36	poisson	Log3	5000	0.84823

The **Random Forest Regression** use R2 value(Best hyper parameter is absolute_error, Sqrt, n_estimators=100 = 0.8495)

6.) Mention your final model, justify why u have chosen the same.

- My Best Model is Random Forest Regression is giving me better accuracy than other algorithms I used for this dataset and the accuracy r^2 score is 0.8495. But I need to increase this model accuracy to above 90% to save for deployment.

1. Multiple LINEAR REGRESSION(R2 Value) = 0.9358

2.SUPPORT VECTOR MACHINE:

S.no	Hyper Parameter	Linear (r2_score)	RBF (Non Linear r2_score)	Poly (r2_score)	Sigmoid(r2 Score)
1	C0.01	-0.0570	-0.0575	-0.0500	-0.0500
2	C10	-0.0396	-0.0568	-0.0500	-0.0500
3	C100	0.1065	-0.0507	-0.0100	-0.0300
4	C1000	0.7803	0.0068	0.2600	0.1800
5	C2000	0.8767	0.0675	0.4800	0.3900
6	C5000	0.9004	0.2100	0.7900	0.7300

The SVM Regression use R2 value(Linear and hyper parameter (C5000) = 0.9004)

3.DECISION TREE:

S.NO	criterion	MAX_FEATURE	SPLITTER	R VALUE
1	MSE	Auto	Best	0.9087
2	MSE	Auto	Random	0.6642
3	MSE	Sqrt	Random	0.7055
4	MSE	Sqrt	Best	0.4218
5	MSE	Log2	Random	0.7055
6	MSE	Log2	Best	0.4218
7	friedman_mse"	Auto	Best	0.9087
8	friedman_mse"	Auto	Random	0.6642
9	friedman_mse"	Sqrt	Random	0.7055
10	friedman_mse"	Sqrt	Best	0.4218
11	friedman_mse"	Log2	Random	0.7055
12	friedman_mse"	Log2	Best	0.4218

13	absolute_error	Auto	Best	0.9329
14	absolute_error	Auto	Random	0.7275
15	absolute_error	Sqrt	Random	-1.3619
16	absolute_error	Sqrt	Best	0.6033
17	absolute_error	Log2	Random	-1.3619
18	absolute_error	Log2	Best	0.6033
19	poisson	Auto	Best	0.9134
20	poisson	Auto	Random	0.6642
21	poisson	Sqrt	Random	0.7055
22	poisson	Sqrt	Best	0.2589
23	poisson	Log2	Random	0.7055
24	poisson	Log2	Best	0.2589

The **DECISION TREE Regression** use **R2 value**(Best hyper parameter is absolute_Error,Auto,Best = **0.9329**)